AI Embedded HIL

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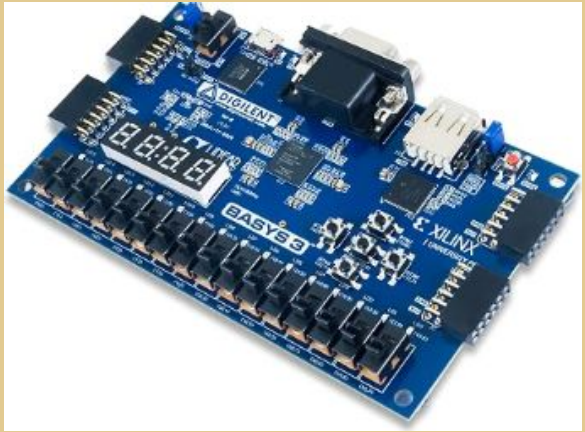


Welcome to the 2nd meeting!

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Hardware is here!

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New board = more installing...

- Installation instructions
- For those interested in the board: Digilent Basys 3 Artix-7
- This will also take a while, so do this during/after the meeting

Subteams

Hardware Team

- FPGA design (trace capture + UART decoder in SystemVerilog)
- Pi firmware (bootloader and test programs)
- Wiring & bring-up (connect the Pis to the FPGA)

Software Team

- AI agent loop (generate, test, and fix firmware)
- Runner & backend (compile, flash, and queue jobs)
- Web dashboard (live agent and trace views)

Choose Subteam



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Research Time

Hardware Team

- FPGA read/send samples
- FPGA and Raspberry Pi pin mapping
- Raspberry Pi bootloading

Software Team

- Prompting: what does the agent need to know?
- Automatically deploy/reset the Pi
- UI to send prompts/watch the agent's progress

References

- *Project inspired by: [Sentinel] (read this!)*

- FPGA Board: [Amazon]
- Raspberry Pis: [Tinkersphere]
- USB-to-UART adapter: [Shillehtek]
- MicroSD Card: [Amazon]

Attendance

Boring Club HIL Attendance Form - Google Forms



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Thanks for coming!

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